Supply Chain & Delivery Performance Analysis

1. Project Overview

This project analyzes the delivery performance of DataCo Global, a worldwide distribution company, using a supply chain dataset of 180,519 orders across multiple markets and product categories. The goal is to uncover the drivers of late deliveries, determine whether poor delivery reliability is systemic or localized, and assess how delivery performance relates to profitability — in order to guide operational and strategic business decisions.

2. Dataset Summary

- **Rows:** 180,519
- **Columns:** 53 (reduced after cleaning)
- **Key Features:**
  - Order & delivery details (order date, shipping date, days for real vs scheduled shipment, delivery status, late delivery risk).
  - Shipping information (shipping mode, and a derived service tier: Priority, Standard, Economy).
  - Geography (market, order region, order country).
  - Product & financials (category, sales, order profit per order).
  - Data quality: mixed date/text formatting in date columns; redundant and PII columns (email, password, names, product image, product description); zero missing values in key fields.

3. Data Cleaning & Preparation using Excel and Power Query

We began with data profiling and cleaning before any analysis:

- **Data Profiling:** Reviewed all 53 columns and logged data-quality issues (encoding, data types, completeness) prior to making changes.
- **Date Standardization:** The order date and shipping date columns were stored inconsistently — partly as text and partly as mismatched date formats. Both were re-typed into a single consistent datetime format using Power Query (Using Locale, English US).
- **Missing Data Check:** Verified zero missing values across key fields (order date, shipping date, sales, profit, customer id, region) using COUNTBLANK.
- **Column Removal:** Dropped six redundant / PII columns — customer email, password, first and last name, product image, and the empty product description.

- **Data Consistency Check:** Confirmed row-level uniqueness of Order Item Id and retained Order Zipcode blanks (expected for non-US orders).
- **Feature Enrichment (VLOOKUP):** Created a Shipping Mode → Service Tier lookup and mapped it onto every order with an exact-match VLOOKUP.

4. Exploratory Analysis using Pivot Tables

We used pivot tables to sanity-check the data and surface early insights:

1. **Late Delivery Rate by Shipping Mode** – Compared the late-delivery rate across the four shipping modes.

Shipping Mode	Late Delivery Rate
First Class	95%
Second Class	77%
Same Day	46%
Standard Class	38%

2. **Late Delivery Rate by Region** – Compared late-delivery rates across all 23 order regions; rates were uniform (49%–58%), clustered around the 55% overall average.
3. **Profit by Shipping Mode** – Compared total and average profit per order across modes.

Shipping Mode	Total Profit	Avg Profit / Order
Standard Class	$2,370,454	$22.00
Second Class	$750,308	$21.31
First Class	$643,122	$23.12
Same Day	$203,018	$20.85

5. Dashboard in Power BI

We built an interactive dashboard in Power BI, driven by seven DAX measures (late delivery rate, on-time rate, average delivery delay, total sales, total profit, average profit per order, and total orders). The dashboard presents insights visually and allows filtering by market.

The dashboard includes:

- KPI cards – Late Delivery Rate (54.83%), Total Profit ($3.97M), Total Orders (181K), Total Sales ($36.78M).
- Late Delivery Rate by Shipping Mode – highlighting First Class as the worst performer.
- Total Profit by Product Category – showing profit concentration (Pareto pattern).

- Late Delivery Rate by Service Tier – a business-friendly rollup.
- Late Delivery Rate Trend (2015–2018) – showing a persistent ~55% rate.
- Order Volume by Country – a geographic map of order distribution.
- Market slicer – for interactive filtering.

6. Key Findings

- **Premium modes are least reliable:** First Class is late 95% of the time versus 38% for Standard Class — the opposite of what customers would expect from premium shipping.
- **The problem is systemic, not geographic:** Late rates are uniform (~55%) across all regions, ruling out location as the cause.
- **No margin trade-off:** Average profit per order is nearly identical (~$22) across all modes, so premium shipping's poor reliability carries no offsetting benefit.
- **Profit is concentrated:** A few categories (Fishing, Cleats, Camping) drive most profit.
- **The issue is chronic:** The ~55% late rate persists across 2015–2018 with no improvement.

7. Business Recommendations

- **Audit Premium Delivery Windows** – A 95% late rate on First Class suggests promised timelines may be unrealistic; re-baseline them against achievable times.
- **Treat It as a Process Problem** – Because lateness is uniform across regions, target the fulfilment and scheduling process rather than specific geographies.
- **Steer Volume to Standard Class** – With equal margin but far better reliability, route non-urgent orders to Standard.
- **Protect Top Profit Categories** – Prioritise stock and delivery reliability for the leading categories (Fishing, Cleats, Camping).
- **Establish Ongoing Monitoring** – Make the dashboard a standing operational report so the problem is tracked to resolution.